

JEE MAIN REVISION MASTER: CENTER OF MASS AND COLLISION

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 8 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Center of Mass of Two-Particle & Continuous Systems	8 PYQs	Center of Mass

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Center of Mass of Two-Particle & Continuous Systems	$X_{cm} = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}, \quad r_1 = \frac{m_2}{m_1 + m_2} d, \quad r_2 = \frac{m_1}{m_1 + m_2} d$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Center of Mass of Two-Particle & Continuous Systems (High Yield — 8 PYQs)

Core Principle: Position of center of mass $R_{cm} = (\sum m_i r_i / \sum m_i)$. For a two-particle system, the center of mass divides the line joining them inversely as their masses ($m_1 r_1 = m_2 r_2$).

■ **Caution / JEE Pitfall:** Center of mass is closer to the heavier mass, not the geometric center.

■ **Speed Trick:** If no external horizontal force acts on a system ($F_{ext} = 0$), the position of the center of mass remains completely unchanged even if internal parts move!

Representative Source PYQs:

- [2025 Shift] Q#1 (MCQ): "During an elastic collision between two bodies, which of the following statements are correct? I. The initial kinetic energy is equal to the..." [Ans: N/A]
- [2025 Shift] Q#2 (MCQ): "Three particles of mass 1kg, 2kg and 3 kg are placed at the vertices A, B and C respectively of an equilateral triangle ABC of side 1 m. T..." [Ans: N/A]
- [2022 Shift] Q#3 (MCQ): "The centre of mass of an extended body on the surface of the earth and its centre of gravity..." [Ans: C]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2025 (Q#1)

During an elastic collision between two bodies, which of the following statements are correct? I. The initial kinetic energy is equal to the final kinetic energy of the system. II. The linear momentum is conserved. III. The kinetic energy during Δt (the collision time) is not conserved.

- (A) II and III only
- (B) I and III only
- (C) I, II and III
- (D) I and II only

Correct Answer: None

Solution Analysis: I. Final total kinetic energy equals initial II. Linear momentum is always conserved III. During collision kinetic energy gets partly converted to potential energy $\therefore$ All statements are correct